A Prerequisite Attention Model for Knowledge Proficiency Diagnosis of Students

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ABSTRACT

With the rapid development of intelligent education platforms, how to enhance the performance of diagnosing students' knowledge proficiency has become an important issue, e.g., by incorporating the prerequisite relation of knowledge concepts. Unfortunately, the differentiated influence from different predecessor concepts to successor concepts is still underexplored in existing approaches. To this end, we propose a **P**rerequisite **A**ttention model for **K**nowledge **P**roficiency diagnosis of students (PAKP) to learn the attentive weights of precursor concepts on successor concepts and model it for inferring the knowledge proficiency. Specifically, given the student response records and knowledge prerequisite graph, we design an embedding layer to output the representations of students, exercises, and concepts.Influence coefficient among concepts is calculated via an efficient attention mechanism in a fusion layer. Finally, the performance of each student is predicted based on the mined student and exercise factors. Extensive experiments on real-data sets demonstrate that PAKP exhibits great efficiency and interpretability advantages without accuracy loss.

CCS CONCEPTS

• **Information systems** → *Data mining*; • **Social and professional topics** → *Student assessment*.

KEYWORDS

Cognitive diagnosis, Attention mechanism, Intelligent education

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1 INTRODUCTION

With the accumulation of rich student learning data stored in massive online education platforms [9], how to diagnose the knowledge proficiency (i.e., mastery level of different knowledge concepts) of each student has become an important issue. Figure 1 shows a concise example of this research task. Here, a student learnt some knowledge concepts by practicing on the related exercises and getting the response Based on response records and the relations between exercises and concepts, this task can output the student's mastery level of different concepts. In fact, the diagnosis results can benefit many educational applications, such as learning path planning [11] and course recommendation [18].

While considerable efforts have been made for knowledge proficiency diagnosis of students, a long-standing research challenge is how to effectively incorporate the influence of knowledge structure for inferring the mastery level of knowledge concepts [10]. Especially for prerequisite relation illustrated in Figure 3(a), the knowledge mastery distribution is influenced unidirectionally from predecessor to successor via directed dependencies [7]. Recently, some approaches tried to improve the diagnosis precision by incorporating the knowledge structure, e.g., prerequisite relation [7, 15, 21]. However, existing approaches usually model the influence of each concept pair with prerequisite relation equally. For example, in [1], all the predecessor-successor pairs are modeled as regularization with the same coefficient. Actually, the influence degree among concepts may be different according to the analysis result illustrated in Figure 3(b). Specifically, we clearly observe that one successor concept is only highly correlated with part precursor concepts (i.e.,

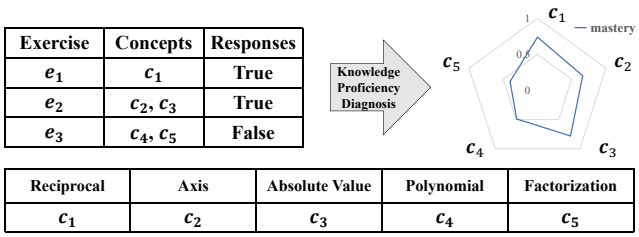

Figure 1: A toy example of knowledge proficiency diagnosis.

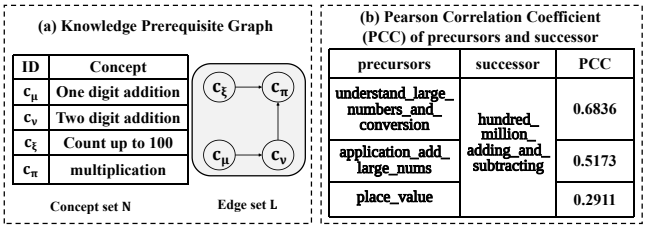

Figure 2: (a) A concise example of knowledge prerequisite graph. (b) The illustration of differentiated correlation of multiple predecessor concepts with same successor concepts.

high PCC) while weak correlated with others based on students' exercising response. Unfortunately, the influence degrees among concepts are not given in advance, how to measure them is a challenging and non-trivial task.

To this end, in this paper, we propose a **P**rerequisite **A**ttention model for **K**nowledge **P**roficiency diagnosis of students (PAKP), with the aim of enhancing the performance through exploring the differentiated influence between concepts. To be specific, we attempt to explicitly learn the influence coefficient of predecessor concepts over successor concepts via attention mechanism and infer the knowledge proficiency by linearly integrating these multiple-predecessor influence. Experiments on real-world datasets demonstrate the high efficiency and interpretation of our PAKP model.

2 RELATED WORK

Knowledge Proficiency Diagnosis. Existing approaches can be divided into traditional models such as IRT [5], MIRT [14] and DINA [3], and deep-learning based models such as DIRT [2], NCD [19], IRRCD [16], CDGK[21] and RCD [7]. Some of them tried to incorporate the dependency relations among knowledge concepts for enhancing the diagnosis performance. For example, a recent model RCD [7] achieved better performance by comprehensively modeling the diverse relations such as student-exercise interactions and exercise-knowledge relations. However, few of them explore the diverse influence from predecessor to successor concepts upon the prerequisite relation. Therefore, we attempt to explore this issue in an efficient and interpretable way.

Prerequisite Relation. The prerequisite relation among knowledge concepts was first proposed in [6], which means that before a student aims to learn subsequent concepts, he must achieve a level of mastery in prerequisite concept [20]. Recently, it has been widely used in many education scenarios, such as curriculum planning [22], cognitive diagnosis [7, 21], and so on [4, 12]. In this paper, exploring the influence mechanism among concepts will also promote the research and applications of prerequisite relation.

3 PRELIMINARIES

Knowledge Prerequisite Graph. The prerequisite relation among concepts is modeled as a graph denoted by $G = (C, L)$, called knowledge prerequisite graph. Here $C = \{c_1, c_2, \cdots, c_K\}$ is the set of all K knowledge concept nodes, and $L = \{(c_i, c_j) | c_i, c_j \in C, \text{ and } i \neq j\}$ is the set of edges. A network G can be also represented by the adjacent matrix $A = (A_{ij} \in \{0, 1\})^{K \times K}$. As shown in Figure 3(a), $A_{ij} = 1$ indicates that concept c_i is precursor of the concept c_j, and

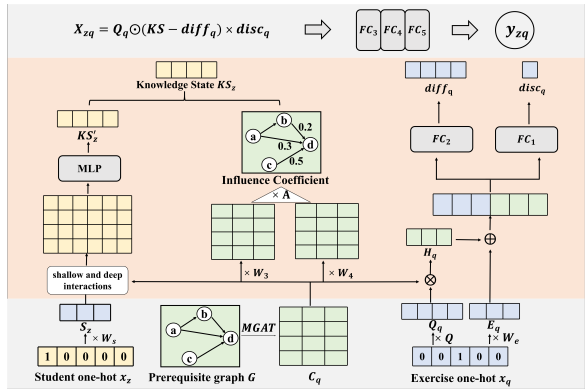

Figure 3: The PAKP framework. From bottom to top are embedding layer, fusion layer, and diagnosis layer respectively.

$A_{ij} = 0$ otherwise. We additionally define an influence coefficient matrix $\mathcal{W}_1$, which is used to measure the differentiated influence among concepts and learned through model training.

Problem Definition. Given students' response record R, exercise-concept correlation matrix Q, and knowledge prerequisite graph G, the goal of our task is to diagnose students' knowledge proficiency via exploring the differentiated influence between precursor concepts and successor concepts in prerequisite relation. Suppose there are N students, M exercises, and K knowledge concepts represented as $S = \{s_1, s_2, ..., s_N\}$, $E = \{e_1, e_2, ..., s_M\}$, and $C = \{c_1, c_2, ..., c_K\}$, respectively. In addition, there is an exercise-concept correlation matrix $Q = (Q_{ij} \in \{0, 1\})^{M \times K}$, where $Q_{ij} = 1$ means the exercise e_i is related to the knowledge concept c_j and $Q_{ij} = 0$ otherwise. We employ a set of triplet (s_z, e_q, r_{zq}) to denote the response logs R, where $s_z \in S$, $e_q \in E$, and $r_{zq} \in \{0, 1\}$ is the score student s_z got on exercise e_q. Here $r_{zq} = 1/0$ means the answer is correct/wrong.

4 MODELS OF PAKP

4.1 Embedding Layer

The embedding of the student denoted by one-hot representation $x_z \in \mathbb{R}^{1 \times N}$ can be obtained by $S_z = x_z \times W_s$, $W_s \in \mathbb{R}^{N \times d}$, and the exercise, denoted by the one-hot representation $x_q \in \mathbb{R}^{1 \times M}$, is also embedded by $E_q = x_q \times W_e$, $W_e \in \mathbb{R}^{M \times d}$, where W_s and W_e are the trainable parameter matrices, and d is the embedding dimension.

For the concepts, to capture the structural information among knowledge concepts, we update them in G by aggregating neighbor

information, the embedding of concepts Q_q can be obtained by

$$H_q = Q_q \times C_q,\ C_q = MGAT(C^0, G),\ C_q, C^0 \in \mathbb{R}^{K\times d}, \quad (1)$$

where $Q_q \in \mathbb{R}^{1\times K}$ is the binary representation obtained from the Q matrix, $MGAT$ is a multi-layer GAT network [17] and C^0 is the initial concept features.

4.2 Fusion Layer

The fusion layer outputs the student and exercise factors for the diagnosis, where the student factor incorporates knowledge prerequisite relations in the influence coefficient matrix with his temporary knowledge proficiency, and the exercise factors are obtained by applying two MLPs to the embedding of exercise and concepts.

4.2.1 ***Temporary Knowledge Proficiency.*** To evaluate student's mastery on knowledge concept c_i, only the student's embedding S_z and c_i's embedding $C_q^i, i \in [1, K]$, i.e., the i-th row of C_q, are utilized as the inputs. Inspired by the famous user-item interaction modeling approach DeepFM [8] in the field of recommendation system, we extract both shallow and deep interactions between students and concepts with the following strategy. Specifically, the shallow interaction $h_1 \in \mathbb{R}^{1\times d}$ is obtained via the application of an average operation on the inputs $h_1 = \left(S_z + C_q^i\right)/2$. The deep interaction $h_2 \in \mathbb{R}^{1\times d}$ is extracted based on the multiplicative operation on the inputs $h_2 = S_z \odot C_q^i$. Next, a MLP is employed to extract higher order information from shallow and deep interactions by

$$KS'_{zi} = MLP\left(x^0\right),\ x^0 = h_1 \oplus h_2,\ x^0 \in \mathbb{R}^{1\times 2d}, \quad (2)$$

where $\oplus$ is the concatenate operation, $MLP(\cdot)$ is a MLP mapping a $1\times 2d$ tensor to a scalar value, and $KS'_{zi} \in \mathbb{R}^1$ is a scalar value measuring the student x_z's proficiency without considering knowledge structure. The above procedure is repeated for all knowledge concepts, then the student x_z's overall temporary proficiency vector $KS'_z = (KS'_{z1}, KS'_{z2}, \cdots, KS'_{zK})$ can be obtained.

4.2.2 ***Influence Coefficient Matrix.*** The influence coefficient matrix $\mathcal{W}_1 \in \mathbb{R}^{K\times K}$ is proposed to measure the influence of predecessor concepts over successor concepts, where $\mathcal{W}_1^{ij}$ measures the influence degree of the predecessor concept c_i on successor concept c_j. Higher value of $\mathcal{W}_1^{ij}$ indicates the greater influence of predecessor concept c_i on successor concept c_j. Based on self-attention, the proposed influence coefficient matrix is calculated by

$$\mathcal{W}_1 = Softmax(\frac{(C_q \times W_3) \times (C_q \times W_4)^T}{\sqrt{d}} \times A), \quad (3)$$

where the computed matrix $\mathcal{W}_1$ is equivalent to the attention map in self-attention, $W_3, W_4 \in \mathbb{R}^{d\times d}$ are trainable matrices, and A is the adjacent matrix of graph G. Here, A ensures that the value of the coefficient is zero if there is no dependency between two concepts. Here the $Softmax(\cdot)$ operation is used for normalizing the coefficients of each column in the influence coefficient matrix.

4.2.3 ***Knowledge Proficiency.*** To incorporate the differentiated influence of different predecessor concepts over successor concepts, included in $\mathcal{W}_1$, into the student's temporary knowledge proficiency KS'_z, the first step is to consider the influence of the precursor concepts on the successor concepts by $KS_z^{pre} = KS'_z \times \mathcal{W}_1, KS_z^{pre}$, then the obtained $KS_z^{pre} \in \mathbb{R}^{1\times K}$ is linearly summed with KS'_z as $KS_z = \alpha \cdot KS'_z + (1-\alpha) \cdot KS_z^{pre}$. Here $KS_z \in \mathbb{R}^{1\times k}$ is the final knowledge proficiency, and α is a hyper-parameter for balancing KS_z^{pre} and KS'_z. Note that the linear weighted-sum operation and the mined influence coefficient matrix provide good computational efficiency and interpretability.

Table 1: The statistics of the datasets.

Dataset	Students	Exercises	Response Records	Concepts	Prerequisite Relations
Junyi	2,000	835	277,089	835	988
Math	1,967	1,686	118,348	61	41

4.2.4 ***Exercise Factors.*** The exercise factors contains two parts: the difficulty of the exercises $diff_q$ and the discrimination of the exercises $disc_q$. Based on the exercise embedding E_q and concept embedding H_q, the exercise factors can be computed by

$$\text{disc}_q = FC_1(E_q \oplus \frac{H_q}{sum(Q_q)});\ \text{diff}_q = FC_2(E_q \oplus \frac{H_q}{sum(Q_q)}), \quad (4)$$

where $(E_q \oplus \frac{H_q}{sum(Q_q)}) \in \mathbb{R}^{1\times 2d}$, $FC_1(\cdot)$ and $FC_2(\cdot)$ are two fully connected (FC) layers used for mapping two $1\times 2d$ tensors to a scalar value and a $1\times K$ tensor, respectively.

4.3 Diagnosis Layer

In the diagnosis layer, the first layer is formulated as

$$X_{zq} = Q_q \odot \left(KS_z - \text{diff}_q\right) \times \text{disc}_q, \quad (5)$$

where $X_{zq} \in \mathbb{R}^{1\times K}$ is latent representation. Then three FC layers are used to predict student's response on exercise as

$$y_{zq} = FC_5(X_4),\ X_4 = FC_4(X_3),\ X_3 = FC_3(X_{zq}), \quad (6)$$

where y_{zq} is the final output, the dimensions of X_3 and X_4 are determined by $FC_3(\cdot)$ and $FC_4(\cdot)$. After training the PAKP by minimizing the cross entropy loss between the output y_{zq} and label r_{zq}, the diagnosis result KS_z can be obtained.

5 EXPERIMENTS

5.1 Experimental Settings

5.1.1 ***Dataset Descriptions.*** Two real-world datasets Junyi[13] and Math are used to verify the effectiveness of our proposed PAKP, where Junyi is a public dataset crawled from the platform *Junyi Academy*, and Math is another private dataset collected by an online education corporation. Table 1 summarizes their statistics.

5.1.2 ***Comparison Approaches.*** Six state-of-the-art diagnosis approaches used to be compared with our PAKP method, including MIRT[14], DINA[3], NCD[19], IRRCD[16], CDGK[21] and RCD[7].

5.1.3 ***Parameter Settings.*** For fair comparison, the parameter settings for all compared approaches are the same as those in their original papers to hold their best performances. In addition, only the students whose response records' length is more than 15 will be used for experiments, and all remaining students in each dataset are randomly split into 70%/10%/20% for training/validation/testing. In our PAKP, the embedding dimension d is set to 128, the output dimension of $FC_3(\cdot)$, $FC_4(\cdot)$, and $FC_5(\cdot)$ are set to 258, 128, and 1, respectively. The PAKP model is trained by the Adam with the

Table 2: Performance of baseline models and our PAKP on all datasets on predicting student responses.

Datasets	Math			Junyi		
Metrics	ACC	RMSE	AUC	ACC	RMSE	AUC
MIRT	72.21%	43.92%	73.54%	78.94%	39.33%	70.40%
DINA	69.74%	46.22%	72.67%	58.95%	47.40%	64.18%
DINA-IRR	62.97%	49.84%	65.62%	52.64%	49.68%	50.04%
NCD	74.13%	41.51%	77.19%	80.70%	37.96%	72.75%
CDGK	74.18%	41.58%	77.35%	80.98%	37.90%	72.83%
RCD	75.27%	40.41%	79.27%	82.34%	**36.35**%	76.32%
PAKP	**75.40**%	**40.39**%	**79.29**%	**82.42**%	36.38%	**76.34**%

batch size 32 and learning rate 0.0002. α and β are set to 0.3 and 0.7 to hold the best performance. All experiments are implemented with *PyTorch* and executed under Tesla P100 GPU.

5.2 Experimental Results

5.2.1 ***Student Performance Prediction.*** In this part, we will validate the effectiveness of our proposed PAKP by comparing it with 6 state-of-the-art diagnosis approaches. Table 2 summarizes their comparison results in terms of accuracy (ACC), root mean squard error (RMSE), and area under cure (AUC). As can be observed, the ACC, RMSE, and AUC values of our proposed PAKP are competitive to those of RCD and significantly better than those of other compared approaches. In spite of the comparable performance to our PAKP, RCD employs graph embedding technologies on a multi-layer graph with huge number of nodes, which leads to an extremely high computational complexity. Instead of only focusing on improving the accuracy, our PAKP also pays attention to good computational efficiency. Table 3 (Left) compares the time cost by each batch training (uniformly set to 32) in the training of PAKP and RCD, it can be seen that each batch training of PAKP only takes less than 1/30 and 1/140 cost time of each batch training of RCD, but there is no performance loss in our PAKP. To sum up, the effectiveness and efficiency of our proposed PAKP can be validated.

5.2.2 ***Influence Coefficient Quality.*** This part will present the quality of the mined influence coefficients. Specifically, we use the PCC computed from the students' records as the true influence coefficients, and two widely adopted ranking metrics MRR and NDCG are used to measure the ranking correlation between mined influence coefficients and true influence coefficients, where higher values of the metrics mean better ranking quality of the mined influence coefficients. As shown in Table 3 (Right), the influence coefficient matrix mined in our proposed PAKP can provide pretty good results that are close to the true rankings.

5.2.3 ***Ablation Study.*** To verify the effectiveness of the influence coefficient matrix for cognitive diagnosis, we further compare PAKP with KP on Math and Junyi, where KP is a variant of PAKP that removes the influence coefficient matrix and directly takes the student's temporary knowledge proficiency as knowledge proficiency, Figure 4 shows the comparison results of PAKP and KP in terms of ACC, RMSE, and AUC, it can be observed that the performance of PAKP is significantly better than KP on two datasets, which indicates that the proposed influence coefficient matrix is effective for our PAKP to diagnose knowledge proficiency.

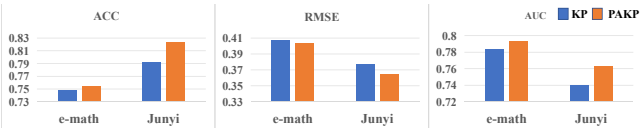

Figure 4: Ablation experiment results.

Table 3: Left: each batch training time of RCD and PAKP; Right: the quality of the mined influence coefficient matrix.

Methods	Math	Junyi	MRR	NDCG@1	NDCG@2	NDCG@3
RCD (PAKP)	0.98s (0.03s)	5.69s (0.04s)	0.6515	0.7197	0.8496	0.8945

5.2.4 ***Case Study.*** For deeper insight into the influence coefficient matrix mined in our proposed PAKP, Figure 5 takes a student chosen from Junyi dataset as an illustrative case, which shows the prerequisite relations and the influence coefficients among concepts A, B, and C, and gives the student's response logs and mastery proficiency on each concept. It can be seen that our PAKP diagnoses this student with a poor mastery level on A while given a high mastery level on B, which keeps consistent with the student's response logs. In addition, the higher influence coefficient between B and C than that between A and C indicates that B plays a more important role than A in the mastery assessment of C, thus the diagnosed mastery on C is also at a high level. In summary, the mined influence the coefficient matrix as well as the linear fusion operation enable our proposed PAKP to hold good interpretability for the diagnosis.

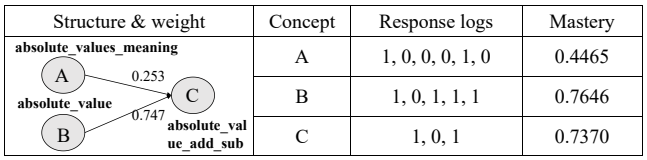

Figure 5: Diagnosis example of a student in Junyi.

6 CONCLUSION

In this paper, we proposed a prerequisite attention model for student knowledge proficiency diagnosis by capturing the differentiated influence of different predecessor concepts over successor concepts. Specifically, the embedding of exercises, students, and concepts are obtained by an embedding layer. Then, in the fusion layer, the influence coefficient among concepts is learnt via an efficient attention mechanism and students' knowledge proficiency is inferred by linearly integrating these multiple-predecessor influences with high efficiency and interpretation. Finally, the performance of student-exercise interaction is predicted in the diagnosis layer. Experiments on two datasets show the effectiveness of our proposed approach.

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